

Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises

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1 Introduction

This section presents the background of the study, which establishes the context of the research. This section also explains why this particular capstone project is important.

1.1 Project Context

The electronics service industry continues to evolve with the rise of digital tools that enhance customer engagement and streamline business operations. Globally, consumer demand for smart technologies and automation has driven continuous growth in the electronics sector [1]. In the Philippines, the electronics industry remains a vital contributor to national economic performance, accounting for more than half of the country's total exports as of 2016 [2]. Alongside this growth, customer service expectations have also shifted—modern consumers increasingly prefer self-service options that allow them to resolve issues independently before seeking professional help.

In fact, a study revealed that 81% of customers across industries attempt to troubleshoot problems on their own using online resources before contacting service representatives, demonstrating a strong demand for Do-It-Yourself (DIY) support systems [3]. This trend not only reflects customers' desire for convenience, but also helps businesses reduce repetitive service requests and improve operational efficiency. A.D. Soreta Electronics Enterprises, established in 1965 in Sorsogon City, is one of the most reputable electronics service providers in the Bicol Region. It consistently delivers appliance installation, maintenance, and repair services and is an authorized service center for leading brands such as Sharp, Samsung, Haier, Daikin, and Carrier. They also specialize in servicing a wide range of household and office electronic appliances LED TVs, washing machines, refrigerators, air conditioners, audio systems, and other essential home and office equipments. While the company has built a strong reputation for technical expertise and reliability, it still relies on traditional service handling through in-person and phone-based appointments. These manual processes can limit responsiveness and create inefficiencies, especially in cases where customers could potentially resolve minor issues themselves with the right guidance. To address these challenges, the proponents proposed the development of a Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises to enhance customer interaction with A.D. Soreta Electronics Enterprises.

The system will serve two main purposes: first, to showcase the company's full range of services, and second, to provide a structured, knowledge-based Do-It-Yourself (DIY) troubleshooting guide to assist customers in resolving minor appliance issues on their own. This feature is built from pre-collected expert knowledge, including common problems and their step-by-step solutions, tailored to the products serviced by A.D. Soreta Electronics Enterprises. The guide will be presented as a combination of text and images to make instructions clearer and easier to follow. By delivering this information in a user-friendly format, the system empowers customers to perform basic troubleshooting without the immediate need for technician support, enhancing service efficiency and customer satisfaction. If the problem cannot be resolved, the system will guide users to book a service appointment. Bookings can also be rescheduled or cancelled through the platform. Once an appointment is submitted, the system will notify the administrator, who will manually assign a technician based on the nature of the issue and the technician's known expertise.

To ensure fast and flexible development, the system will be built using WordPress, a widely supported content management system. WordPress allows rapid customization and plugin integration for features such as appointment booking, content presentation, and user interaction. The system's effectiveness will be evaluated using the ISO/IEC 25010 software quality model, focusing on functional suitability, usability, and efficiency, which are most relevant to the project's operational and user-oriented goals.

1.2 Purpose and Description

The purpose of this project is to develop a Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises that allows customers to explore the company's services and troubleshoot minor electronic issues through a structured DIY guide. If issues persist after using the guide, users can book, reschedule, or cancel a service appointment directly on the website. The website will also act as a digital hub for showcasing the company's service offerings, including appliance installation, maintenance, and repair. WordPress will be used to develop the platform, ensuring a balance of flexibility, customization, and rapid deployment.

The system's functionalities, content display, user engagement, and appointment, will be developed to address both customer convenience and internal efficiency. The project's success will be evaluated using the ISO/IEC 25010 software quality model, particularly This system is expected to deliver benefits to both the company and its customers. For A.D. Soreta Electronics Enterprises, it reduces reliance on manual service handling, enhances technician assignment coordination, and modernizes the customer service workflow. For customers, it improves access to self-help content and streamlines the booking process, resulting in quicker and more convenient service access. Lastly, this project can serve as a reference for future developers or researchers exploring digital transformation in service-oriented businesses with similar operational requirements.

1.3 Objectives of the Study

This capstone project aims to design and develop a Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D Soreta Electronics Enterprises, with the goal of enhancing customer support and streamlining service appointment management. The specific objectives are:

1.To develop a responsive website as the system's main interface, which includes:

1.1 A company profile section showcasing Soreta Electronics Enterprises' services and contact information.

1.2 A content management module for updating service information and announcements.

1.3 A user authentication system for secure customer access.

2.To implement a Knowledge-based DIY Troubleshooting Guide composed of:

2.1 A decision-tree logic for guiding users through step-by-step troubleshooting

processes.

2.2 A knowledge base containing frequently encountered issues, suggested fixes, and preventive tips.

2.3 A feedback module for user to rate the usefulness of the guide.

3.To create a customer appointment management system with features for:

3.1 Booking new appointments with service type, date, and time selection.

3.2 Rescheduling existing appointments.

3.3 Canceling appointments with optional input.

3.4 Notification module (web-based alerts) for appointment confirmations and reminders.

4.To evaluate the developed system using the ISO/IEC 25010 software quality model, focusing on:

4.1 Functional Suitability

4.2 Usability

4.3 Efficiency

1.4 Scope and Limitations of the study

This project focuses on the development of Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises. The primary function of the system is to provide knowledge-based support through a structured DIY troubleshooting guide that will assist users in resolving minor electronic issues independently. The troubleshooting guide will consist of both text and images to help customers understand and follow basic steps in identifying and possibly fixing common problems with their electronic appliances. If the user is unable to resolve the issue on their own, the system will then offer or encourage the customer to book a service appointment for professional assistance. The basic appointment system will allow customers to book, reschedule, or cancel appointments online. Once a booking is made, the administrator will receive a notification through the system. Since the administrator is already familiar with each technician's individual skillset, they will manually assign the most suitable technician to handle the specific issue reported by the customer. The website will also serve as a digital platform to showcase the complete range of services offered by the company, including appliance installation, repair, and maintenance.

Only two user roles will interact with the system: the customer and the administrator. The customer will have access to the public-facing side of the website, where they can browse service offerings, access the DIY troubleshooting guide, and manage their appointments. Their interaction is limited to viewing content and submitting, rescheduling, or canceling appointments. On the other hand, the administrator will have access to the system's backend panel. From there, the administrator can view all incoming appointment records, manage bookings, and assign technicians manually based on the nature of each service request. The technician, however, is not a system user and will not access or interact with the system in any way. All technician-related tasks are handled outside the system by the administrator.

While the system aims to enhance customer support and improve the overall service process, it is limited in several ways. The technician assignment process will not be automated and will rely solely on the administrator's judgment and experience. The DIY troubleshooting guide will only cover basic and frequently encountered appliance issues and will not extend to complex repair procedures or include interactive or video-based support. Additionally, the current scope does not include the integration of advanced security features or full compliance with data privacy laws due to technical constraints and resource limitations. Features such as customer feedback collection, technician profiles, or real-time chat support are also not part of the system's initial version. Despite these limitations, the system is expected to significantly improve customer experience and operational efficiency by providing structured self-help resources and streamlining the appointment process through a simple, role-based interface.

Section 2

2 Review Related System

The primary aim of this project is to develop a customer-oriented website for A.D. Soreta Electronics Enterprises that provides a Do-It-Yourself (DIY) troubleshooting guide and enables clients to book, reschedule, or cancel appointments. The system is designed to bridge the gap between customer independence and professional support. Through this dual-functionality approach, the project focuses on knowledge-based customer assistance and efficient appointment management. This chapter examines existing systems and technologies relevant to our proposed project, identifying best practices and limitations. The review is organized into three thematic areas: DIY Troubleshooting Guides for

Electronic Appliances, Online Booking and Appointment Systems, and Administrative Platforms for Technician Assignment and Service Management.

2.1 DIY Troubleshooting Guides for Electronic Appliances

The Samsung Smart Tutor application is a remote support tool designed to assist users in troubleshooting minor issues with their devices, including smartphones and appliances. It allows technicians to remotely access the customer's device to diagnose problems and guide users in real time. Although it emphasizes remote support rather than a standalone DIY guide, its usability and user-friendly interface provide a foundation for developing intuitive self-help tools [4]. Similarly, the LG Support app features step-by-step instructions for resolving common appliance issues. The guide is categorized by product type, allowing users to narrow down problems based on symptoms and receive corresponding solutions, enhancing the user experience through clarity and simplicity [5].

Another notable example is the Whirlpool Self Help Portal, which provides interactive troubleshooting tools for various appliances. This system utilizes a question-and-answer flow to help users identify the root cause of issues. Its strength lies in visual cues, such as error code lookups and part identification diagrams, making the process less intimidating for non-technical users [6]. In addition, the Fixya platform serves as a crowdsourced solution center where users post problems and receive community-based answers. Although not brand-specific, its wide coverage and peer-reviewed solutions make it an effective knowledge-sharing tool [7]. Lastly, the Sharp Customer Support Portal offers downloadable manuals, FAQs, and error code decoders for appliances. Its effectiveness is derived from combining multimedia content and text-based instructions, which allows users to engage with different learning styles [8].

2.2 Online Booking and Appointment Systems

One relevant system is the Setmore appointment scheduling platform. It enables users to schedule, reschedule, and cancel appointments with service providers and includes automatic email and SMS reminders. Setmore offers customizable booking pages and time slot management, making it ideal for service-based businesses [9]. Likewise, Booksy is a client-focused appointment booking system that offers a seamless experience for users and providers. It features booking history, rescheduling functions, and real-time availability, which increases operational efficiency [10]. Another system, Acuity Scheduling, allows service providers to customize intake forms, appointment types, and time buffers. It is notable for its synchronization with calendars and payment processors, streamlining the scheduling workflow [11].

Additionally, the Philippine-based application MyKuya offers real-time booking of on-demand services, including electronics repair. It highlights an intuitive interface and prioritizes customer convenience by enabling service bookings through a mobile app. Though focused on urban logistics, its real-time tracking and technician assignment options provide parallels to appointment booking systems for electronics services [12]. Meanwhile, the Zennya app, primarily designed for healthcare appointments, incorporates dynamic rescheduling and cancellation features, which serve as a good reference for implementing a flexible and user-centric appointment module [13].

2.3 Web-Based Systems Developed Using WordPress

WordPress is a widely used content management system (CMS) that powers a significant portion of websites across different industries due to its flexibility, scalability, and availability of plugins. One notable example is the Auto Repair Booking System developed using WordPress and WooCommerce Bookings, which allows customers to book car repair appointments online. This system takes advantage of WordPress’s plugin ecosystem to integrate calendar scheduling, service categorization, and form customization, providing a streamlined experience for both customers and administrators [14]. This model parallels the proposed system’s appointment functionality, where customers can book and manage repair schedules through a user-friendly interface.

Similarly, the Medical Clinic Website developed with WordPress utilizes the Bookly plugin to facilitate patient appointment booking, rescheduling, and cancellation. The system enables administrators to manage patient records, assign doctors manually, and control time slots. This implementation demonstrates how WordPress can support role-based access and controlled scheduling without the need for custom-built back-end systems [15]. These features are relevant to A.D. Soreta Electronics Enterprises, where the administrator oversees technician assignments without requiring technician access to the system. Another example is the Home Services Website that offers on-demand plumbing, electrical, and appliance repair services, all built on WordPress with the WPForms plugin and a custom booking plugin. It uses a categorized troubleshooting FAQ section for self-help before prompting users to book a service. This approach closely aligns with the dual nature of the proposed system—offering a DIY troubleshooting guide followed by appointment functionality [16].

In addition, the Online Appliance Repair Service Portal uses WordPress and the Elementor plugin to design an intuitive layout that includes a service catalog, user registration, and appointment scheduling form. It offers static DIY content with step-by-step visuals, guiding users through basic repair steps for common issues before escalating to professional help. This design strategy supports A.D. Soreta’s goal

of combining knowledge-based troubleshooting with online booking [17]. Finally, the Technical Support Website for Electronics Repair utilizes WordPress with LearnDash and WP Support Plus to build a knowledge base, DIY tutorials, and service request forms. The knowledge base includes articles and image-enhanced guides categorized by appliance type (e.g., refrigerator, washing machine, TV), helping users resolve minor issues independently. This system shows how WordPress can be leveraged not only for content display but also for guided troubleshooting interactions [18].

The use of WordPress in these systems demonstrates its effectiveness in creating functional, responsive, and scalable platforms for service-oriented businesses. For the system being developed for A.D. Soreta Electronics Enterprises, WordPress will serve as the development backbone. It will facilitate a modular setup where the DIY troubleshooting guide can be presented using content blocks with embedded images, collapsible FAQ sections, and appliance-specific categories. Meanwhile, appointment functionality will be implemented using a combination of plugins such as WPForms (for service request submission), WP Simple Booking Calendar (for time slot visibility), and custom admin dashboards to support manual technician assignment by the administrator. This approach allows rapid development without compromising usability or administrative control.

Section 3

3 Technical Background

This section provides a comprehensive overview of the technical framework used in the development of the system. It elaborates on the hardware and software tools employed by the developers, the necessary specifications required by users to access the system efficiently, and the definitions of key technical concepts involved in the project. By presenting these elements, the section ensures a clearer understanding of the system's foundation and its operational environment.

3.1 System Development Specification

This section outlines the technical components used by the developers to build the system. It details the specific hardware, software, and service tools that supported the development process. Identifying these specifications is important not only for documentation but also for replicability and troubleshooting during future updates or system scaling.

Hardware Specification

Table 1: Developer’s Hardware Specification

HARDWARE	DESCRIPTION
Processor	Intel Core i5 or AMD Ryzen 5
Memory	8 GB DDR4 RAM (16 GB preferred)
Display	1920 × 1080 resolution (Full HD)
Storage	256 GB SSD
Graphics	8 GB DDR4 RAM (16 GB preferred)
Networking	Wi-Fi 5/6 or Ethernet

The hardware specifications listed in Table 1 support the developers during the project’s development. The system development will utilize a mid-range hardware setup to ensure smooth multitasking and compatibility with WordPress development tools. A processor such as the Intel Core i5 or AMD Ryzen 5 offers efficient performance for handling local servers, plugins, and WordPress operations [19]. A minimum of 8 GB RAM is necessary to support simultaneous browser testing, CMS manipulation, and wireframing, though 16 GB is preferred for optimal performance. A Full HD display ensures the correct visual rendering of UI/UX components. The 256 GB SSD provides high-speed read/write capabilities essential for storing databases, source files, and backups. Since the system is not graphics-intensive, integrated graphics like Intel UHD or AMD Vega suffice. Lastly, stable networking hardware is required for downloading updates, remote backups, and seamless browser interactions.

Software Specification

Table 2: Developer’s Software Specification

SOFTWARE	DESCRIPTION
Browser	Google Chrome v.124.0.6367.63
Operating System	Windows 11 64-bit, x64-based processor
Development Platform	WordPress 6.x (latest stable release)
Database	MySQL 8.0
Web Server	Apache HTTP Server 2.4.56 (via XAMPP or LocalWP)
Code Editor	Visual Studio Code
Diagram Tool	Draw.io
Prototype Tool	Figma
Document Tool	Overleaf (LaTeX)

The software stack selected supports a web-based application developed using WordPress. The browser, Google Chrome, is equipped with developer tools that aid in testing UI elements, inspecting code, and simulating mobile views [20]. The operating system, Windows 11 64-bit, provides compatibility with modern

development environments and is widely supported. WordPress 6.x, as the development platform, offers plugin integration and content management features well-suited for appointment booking and troubleshooting functionalities [21]. Data is managed using MySQL 8.0, a relational database that pairs seamlessly with WordPress. A local server environment is simulated using Apache HTTP Server 2.4.56, commonly bundled via XAMPP or LocalWP, for testing site behavior before deployment.

Visual Studio Code serves as the primary editor for customizing WordPress themes, functions, and styles. System diagrams like the use case and data flow diagrams are created using Draw.io due to its simplicity and UML support. Figma is employed for creating interface prototypes and ensuring responsive design layouts. Lastly, documentation is handled via Overleaf (LaTeX).

Service Specification

The service specification provides a detailed description of the services that will be used.

Table 3: Developer’s Service Specification

SERVICE	PROVIDERS
Email Notification	SMTP via Gmail API / WP Mail SMTP Plugin
API Integration	Google Calendar API (for scheduling reference)
Authentication	WordPress Default Authentication System
Web Hosting	cPanel-based Shared Hosting / WordPress Hosting
Domain Name System	Custom Domain via Namecheap / GoDaddy
Plugin Management	WordPress Plugin Ecosystem (e.g., WPForms, BookingPress)

In the development of the appointment and troubleshooting system for A.D. Soreta Electronics Enterprises, several online services are utilized to support core functionalities such as notifications, scheduling, authentication, and deployment. These services ensure the smooth interaction between system components and enhance the user experience for both administrators and customers.

For email notifications, the system employs the Simple Mail Transfer Protocol (SMTP) integrated through tools like WP Mail SMTP or the Gmail API, allowing WordPress to send reliable and authenticated emails when customers book, cancel, or reschedule appointments. Email notifications are essential in alerting administrators in real time, as native PHP mail functions in WordPress are often restricted by hosting providers and less secure without SMTP support [22]. To handle calendar-based scheduling, the system may optionally utilize the Google Calendar API, which integrates appointment data into a visual calendar that the administrator can reference. While the technician assignment remains manual,

visualizing bookings in a calendar format improves schedule management and reduces the risk of double-booking or missed appointments [23].

Authentication is managed using WordPress’s native authentication system, which includes role-based access control. In this system, two user roles are defined: the customer and the administrator. WordPress’s built-in user management supports secure login, registration, and role-restricted access without needing third-party services like Firebase Authentication. This simplifies the development while maintaining secure access control.

For hosting, the system is deployed on a managed WordPress hosting platform that supports PHP 8.x, MySQL databases, and SSL encryption. This ensures optimal performance, scalability, and security for both development and production environments. WordPress is often hosted on platforms like SiteGround, Bluehost, or Hostinger, which provide control panels, one-click WordPress installs, and database management tools [24]. Additionally, a domain name and DNS service is essential for professional deployment, ensuring that users can access the system via a human-readable domain. Providers such as Namecheap or GoDaddy are commonly used for registering and managing domain names and DNS records.

Lastly, the system heavily relies on the WordPress plugin ecosystem to extend its core features. Plugins such as WPForms (for troubleshooting guides), BookingPress or Amelia (for scheduling), and WP Mail SMTP (for notifications) enable rapid development and customization. These plugins integrate seamlessly into the WordPress environment and allow the system to function without building custom components from scratch [25].

3.2 User System Requirements

This section defines the minimum and recommended system requirements for users who will interact with the system. Specifying these parameters is critical for ensuring that users experience fast response times, and uninterrupted access to the features of the system.

The hardware and software requirements outlined above provide a structured basis for ensuring both customers and administrators can effectively interact with the system. These specifications support the seamless operation of core features such as the DIY troubleshooting guide, appointment management, and administrative controls. The system, being browser-based and platform-independent, does not demand intensive computational power, but it benefits from modern hardware to optimize user experience.

A general-purpose processor is sufficient for running the system’s web functionalities; however, recommending an 11th Gen Intel® Core™ i5-1135G7 or

Table 4: System Requirements

Component	Minimum Requirements	Recommended Requirements
Processor	Any modern processor (dual-core)	Intel® Core™ i5-1135G7 or higher
Memory	8GB of RAM	8GB or higher
Storage	18 GB available space	20GB or more
Display	15 inch 1280x720 resolution	15 inch FHD 1920x1080 resolution
OS	Windows 10 Home 64-bit	Windows 11 Home 64-bit
Browser	Google Chrome v135.0.7049.42	Google Chrome v135.0.7049.42 or newer
Internet Speed	5–10 Mbps	15–20 Mbps

similar offers enhanced responsiveness and multitasking capabilities, especially for administrators managing multiple booking requests and system records [26]. An 8 GB RAM minimum is standard in modern computing and allows users to operate browsers efficiently, handle dynamic content such as troubleshooting videos or forms, and experience minimal delays. Storage requirements are minimal since most data processing occurs server-side; thus, 18 GB of free space suffices. However, 20 GB or more is recommended to accommodate browser cache, temporary files, and other system utilities without affecting performance.

The display requirement of 1280x720 pixels ensures the user interface is properly rendered, but a Full HD resolution (1920x1080) enhances clarity for form details, service status updates, and troubleshooting images, which can impact user satisfaction [27]. A stable internet connection is essential for real-time appointment booking, rescheduling, and notifications. A connection of 5–10 Mbps is adequate for loading standard web content, while 15–20 Mbps is recommended to ensure faster loading times, especially when accessing multimedia-rich troubleshooting guides or simultaneous admin dashboard activities.

The system runs on modern web browsers, with Google Chrome version 135.0.7049.42 or later recommended to ensure compatibility with current HTML5, CSS3, and JavaScript standards. This supports secure and reliable access across various devices, whether desktop or mobile [28]. Finally, compatibility with Windows 10 or Windows 11 ensures the system functions within a stable, secure, and widely used environment, offering long-term support for users.

3.3 Technical Terms

WordPress is a free and open-source content management system (CMS) that lets anyone build and manage websites easily. Started as a blogging platform, the

Wordpress software now helps users launch various website types. The system will be developed using Wordpress a CMS to ensure quick development, customization, and scalability of content like services, guides ,and appointment form.[30]

Responsive Website a website design approach that adjusts the layout and elements of a website based on screen size and device type.System must be accessible on any device mobile, tablet, or desktop making it a responsive website. This enhances usability and customer satisfaction.[31]

DIY Troubleshooting Guidean expertise-based, self-help tool that guides users to resolve common problems without requiring expert help. The platforms includes a structured Do-it- yourself(DIY) section where customers can try to fix a minor issues before booking a technician.[32]

Appointment Scheduling System a digital tool that automates the scheduling process for services, often with features like reminders and calendar syncing. Users can book, cancel, or reschedule service appointment directly on the website, helping streamline customer service operations [33]

Administrator Panel a backend control area where privileged users can manage system data, user activity, and content. And also only the administrator (not the technician) will access the backend panel to manage appointments and assign technicians based on issue types.[34]

Plugin Integration external modules or add-ons that enhance the capabilities of a web platform or CMS. To extend Wordpress functionalities like adding booking forms or images sliders plugins will be integrated without custom development.[35]

Knowledge-based System system that uses a database of facts and rules to simulate expert decision-making and support users. The DIY troubleshooting guide is structured using a pre-defined knowledge, making it a simple form of a knowledge-based system.[36]. In simpler terms, the “knowledge base” is a digital library of service know-how. It draws from actual technician expertise, manuals, and troubleshooting protocols, and organizes this information into decision-tree formats, FAQs, or guided steps. It doesn’t use artificial intelligence or dynamic learning, but rather relies on curated, static content to assist users.

User Roles a set of permission assigned to users to define what they can view or control within the system. The system only includes two user roles: Customer (for front-end task) and Administrator (for back-end management and technician coordination).[37]

Web-based Platform a software application that is hosted on a server and accessed through a web browser over the internet.[38]

Section 4

4 Design and Methodology

The primary objective of this project is to develop the “Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises.” This system is designed to support the company’s operations by enabling customers to independently diagnose and resolve minor issues with their electronic appliances and to book service appointments online. It simplifies service access, reduces unnecessary technician deployment, and enhances the efficiency of service scheduling. The system reflects the actual workflow of the company, in which only the administrator and customer interact with the platform—technician assignment is handled manually by the administrator outside the system. To fully grasp the architecture and functionality of the system, a detailed discussion of its design and methodology is essential.

This chapter serves as the foundation for the methodologies applied in the planning, analysis, and development of the system. It includes architectural and process diagrams, requirement analysis, the development tools and methodologies used, as well as testing strategies. Furthermore, it outlines the responsibilities of the project proponents, the schedule and timeline of development activities, and the budget management required to ensure a structured and successful system implementation.

4.1 Concept

This subchapter introduces the design methodology that supports the development of the Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises. The system is centered on improving customer service through a web-based platform that integrates a knowledge base for minor troubleshooting and a streamlined appointment booking system. The design is tailored to reflect the roles and workflow described in Section 1, where only customers and administrators interact with the system. As a customer-oriented solution, the system ensures user-friendly access to service options, while administrators maintain control over service appointments and technician assignments. The architecture, data flow, and system interactions are all designed to reflect this simplified yet functional dynamic

4.1.1 System Architecture Diagram

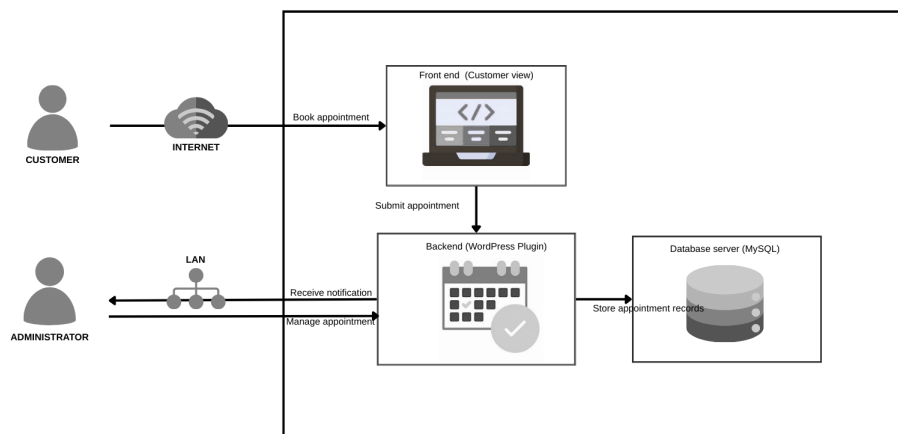


Figure 1: System Architecture Diagram

Architecture diagrams provide a visual overview of how various components of a system interact and communicate with each other. In the case of the Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises, the system is hosted locally using XAMPP, and structured around a three-tier architecture consisting of a presentation layer, application layer, and data layer.

The process begins when a customer accesses the system through a web browser, initiating interaction with the presentation layer, which includes the WordPress frontend pages such as the homepage, services, troubleshooting guide, and appointment form. These components are rendered based on themes and templates provided by the WordPress CMS.

When a customer submits a form—for booking, cancellation, or troubleshooting—the request is passed to the application layer, where WordPress plugins and core functionalities handle form processing, trigger SMTP-based email notifications, and validate user input. The administrator, accessing the backend via the WordPress admin panel, is notified of appointment-related events and can view and manage records.

All data generated or retrieved—such as customer details, troubleshooting inputs,

and appointment schedules—is handled by the data layer, which consists of a MySQL database configured through XAMPP. This database stores all system-related data and ensures reliable backend access for both customer and administrator operations.

4.1.2 Data flow Diagram

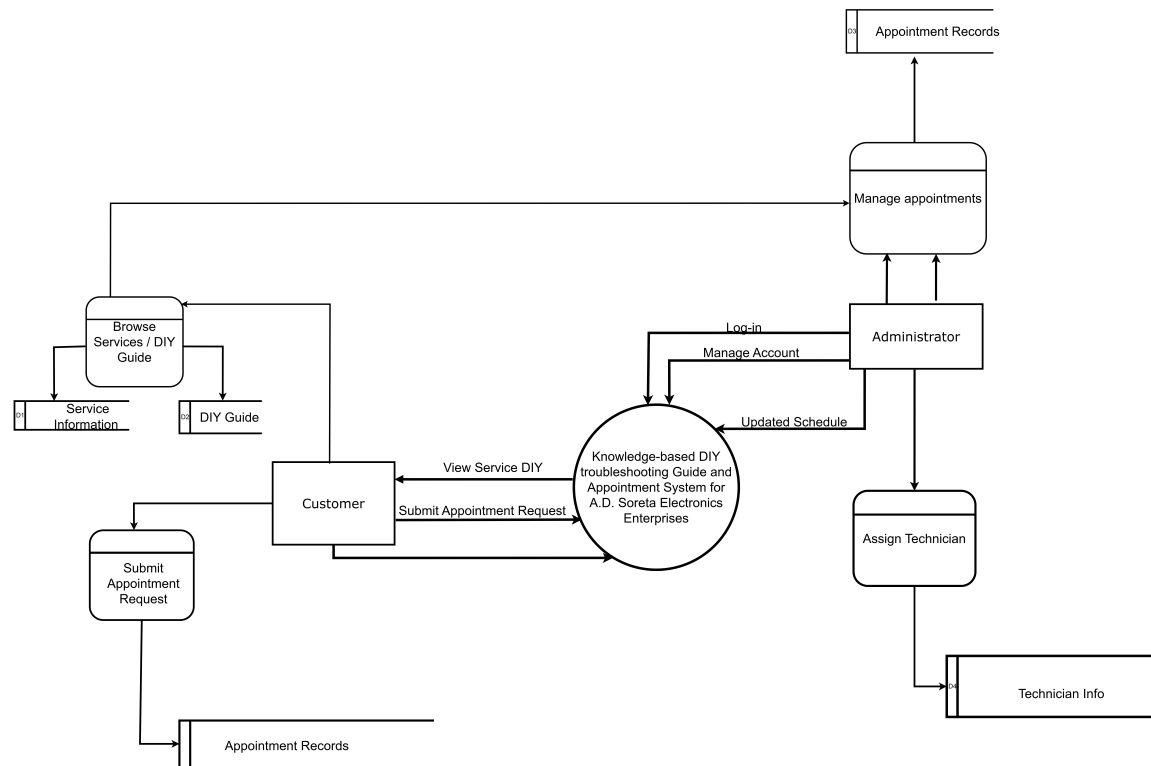


Figure 2: Data Flow Diagram

The system shown in the diagram is the Knowledge-Based DIY Troubleshooting Guide and Appointment System designed for A. D. Soreta Electronics Enterprises. Customers can either solve their problems by themselves with the help of the DIY guides or use it to book service appointments with A. D. Soreta Electronics for their equipment. Customers can search for available services and troubleshooting guides. Through the “Browse Services/DIY Guide” module users can obtain information on available services as well as step-by-step DIY guides for fixing their problem. If they cannot solve the problem with the help of DIY guides, they are required to submit an appointment request, thus via the “Submit Appointment Request” feature, their request is stored in the appointment records and is processed by the administrator. The main system is the portal between the customers, administrators, and the data stores in the system.

Administrators can log in to the system, access account details, and view the schedule of appointments. Through the "Manage Appointments" function administrators can review customer requests, make bookings, and view the current schedule. This schedule is also recorded in the appointment records.

4.1.3 Use Case Diagram

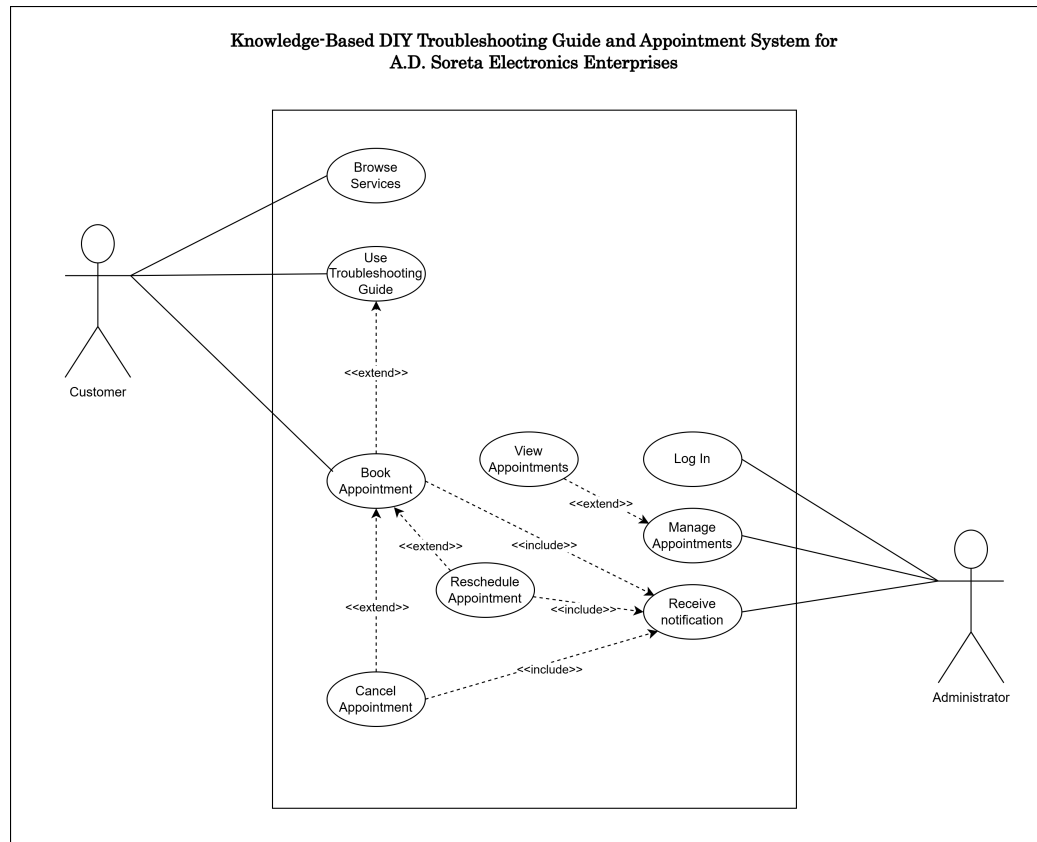


Figure 3: Use Case Diagram

The use case diagram represents a Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises. It involves two main actors: the Customer and the Administrator. The Customer can browse available services, use a troubleshooting guide, and manage their service appointments. Before accessing most features, the Customer must log in, which is a prerequisite for viewing, booking, canceling, or rescheduling appointments.

The troubleshooting guide can be used independently or as part of the appointment process, helping customers diagnose issues before booking a service. Appointment management functions such as canceling or rescheduling are optional extensions of booking an appointment. On the other hand, the Administrator also logs into the

system to perform administrative tasks such as managing all appointments and receiving system notifications related to customer activities. The diagram illustrates how the system supports both user roles in a structured and efficient way, integrating self-help tools with formal service scheduling and management.

4.1.4 UML Activity Diagram

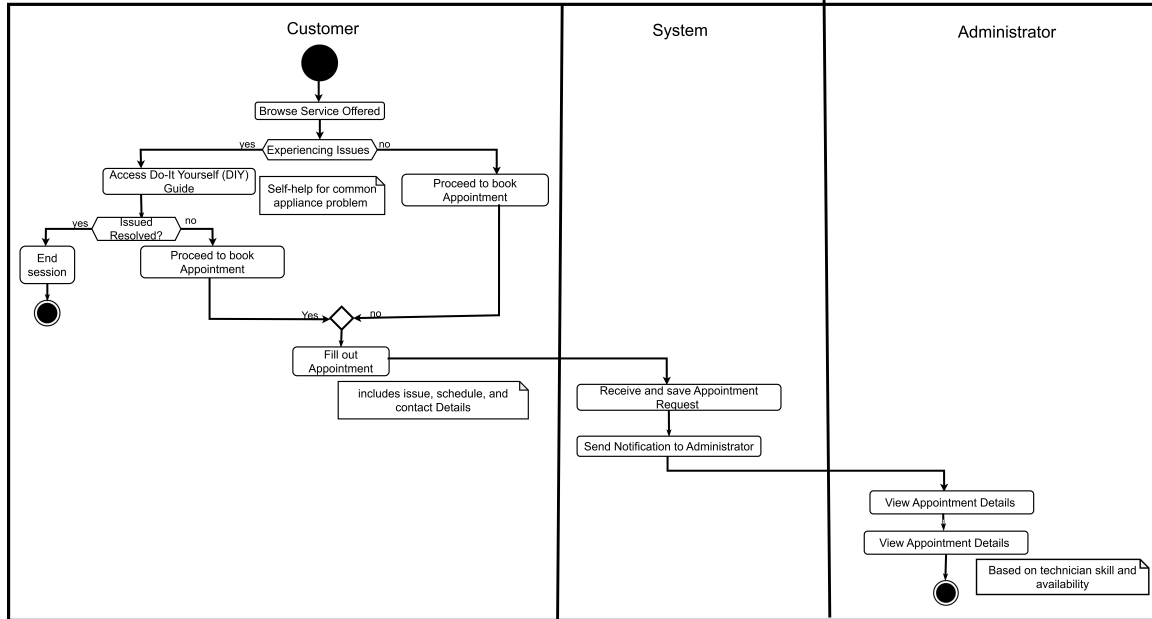


Figure 4: UML Activity Diagram

This activity diagram illustrates the step-by-step process of how the Online Service Management System operates. It all begins with the customer log in, at which point the system verifies their details and confirms them. Next, the customer submits a service request, and the system saves that information, making it visible to the administrator. After reviewing the request, the administrator assigns a technician and updates the schedule accordingly. Finally, the system sends out a notification via SMS or email to keep the customer informed about the update.

4.1.5 Entity Relationship Diagram

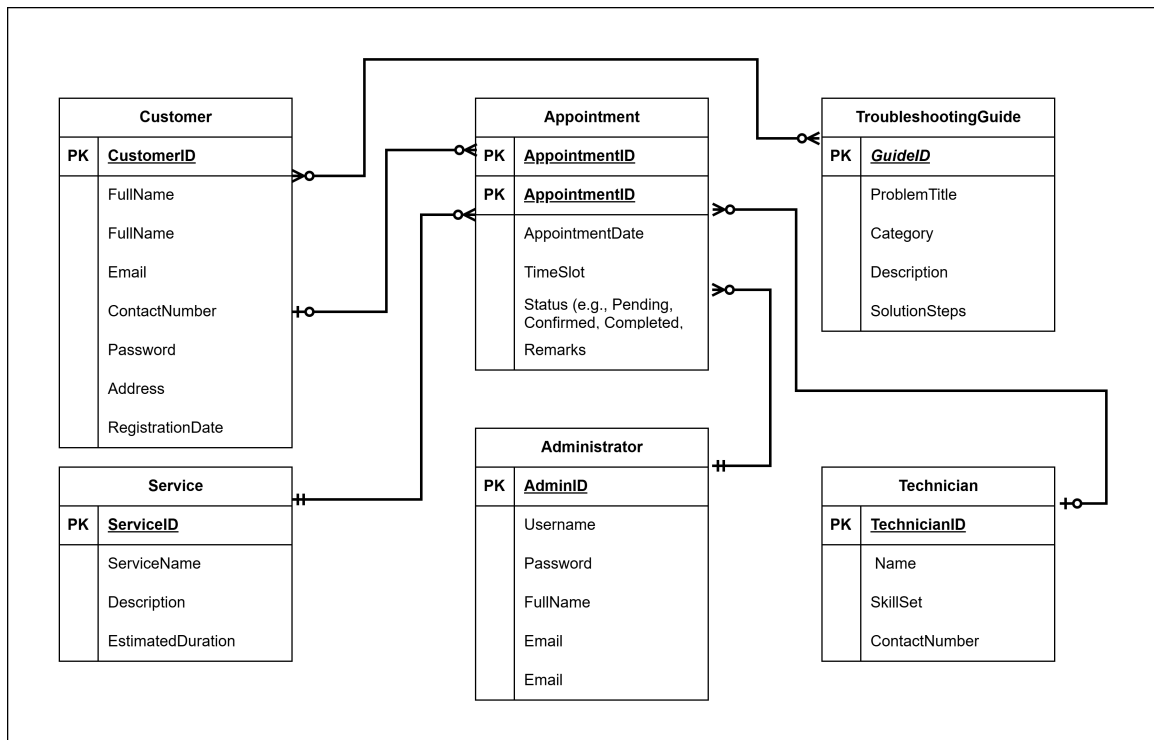


Figure 5: Entity Relationship Diagram

The Customer uses the system to book a service appointment and submits the request to the Administrator. The administrator receives the appointment details, including the selected service and preferred date and time. The administrator then manually assigns a Technician based on the nature of the problem and the technician's expertise, but this assignment process happens outside the system. The administrator also manages the appointment records within the system, updating their status as needed — such as confirming, rescheduling, or canceling appointments. Meanwhile, the Customer can access a set of Troubleshooting Guides available on the website. These guides help the customer try to fix common electronic issues before requesting professional assistance. If the issue persists, the customer proceeds to book an appointment. Although the technician is not a system user and does not interact with the system, their details are stored for reference so the administrator can assign them appropriately. The administrator acts as the central coordinator, handling all system-side processes and maintaining the data needed to ensure efficient service delivery.

4.2 Analysis and Design

This chapter outlines the results of the system analysis and the corresponding design solutions for the development of the Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises. It aims to translate the system’s goals and scope into a set of well-defined technical models and specifications that will guide implementation. The chapter begins with a detailed requirements analysis, covering both functional and non-functional aspects of the system. Following this, system models such as the system architecture, data flow diagram, use case diagrams, activity diagrams, and the entity-relationship diagram (ERD) are presented. These models illustrate how users will interact with the system, how data will flow internally, and how various components will be structured and connected. The objective is to ensure that the proposed system is well-aligned with user needs, efficient in operation, and clear in structure before actual development begins.

4.2.1 Requirement Analysis

This subchapter discusses the system requirements derived from the defined objectives and scope of the project. Requirements are categorized into functional requirements, which describe what the system should do, and non-functional requirements, which define how the system should behave. The functional requirements ensure that all user tasks and system responsibilities are explicitly identified for both customer and administrator roles. On the other hand, the non-functional requirements cover software quality attributes such as usability, performance, and functionality — using the ISO/IEC 25010 quality model as a guideline. These requirements form the foundation of the system design and are critical to developing a solution that is both useful and user-friendly.

4.2.1.1 Functional Requirements

Functional requirements are capabilities of the system that developers must implement to enable users to accomplish their tasks[34]. The functional requirements of the system shown in Table 4.2.1.1 specify tasks of the system for different users within the system.

Table 5: Functional Requirements

Requirement Description	User Role
The system should allow the administrator to view all the functionalities of the system, including service requests, technician assignments, and record management.	Administrator
The administrator should be able to manage and monitor the scheduled appointments of the clients.	Appointment Managements
The system should allow customers to receive SMS notifications and emails regarding the services offered.	Clients

4.2.1.2 Non-functional Requirements

The non-functional requirements, which describe the desired performance and behavior of the system [36]. Table 5 contains a summary of these needs, complete with task references and detailed explanations. The references are based on the Software Quality Standards of the International Organization for Standardization (ISO/IEC 2023).

Table 6: Non-Functional Requirements

Task Requirements	Task Reference
The system should ensure it effectively delivers accurate, complete, and reliable service management functions, aligning with user needs and operational expectations.	Functional Suitability
The system should optimize performance by maximizing resource use and guaranteeing fast reaction times, supporting efficient and seamless operation under a range of workloads.	Efficiency
The system should provide an intuitive, easy-to-navigate, and accessible interface for all users.	Usability

4.3 Development Model

Rapid Application Development (RAD) is a software development methodology that emphasizes iterative prototyping and user feedback to accelerate delivery and ensure alignment with user requirements [29]. Rapid Application Development (RAD) was the chosen development model for the creation of Knowledge-Based DIY Troubleshooting Guide and Appointment System for A.D. Soreta Electronics Enterprises. RAD is a software development methodology that

emphasizes iterative prototyping and fast user feedback, allowing for accelerated delivery of functional systems.

This model is particularly well-suited for the project, given the capabilities of the WordPress platform, which supports rapid deployment, plugin-based feature integration, and flexible customization. By adopting RAD, the development team was able to continuously incorporate client feedback throughout the development process, ensuring that the final system aligned with the actual needs of the business and its customers.

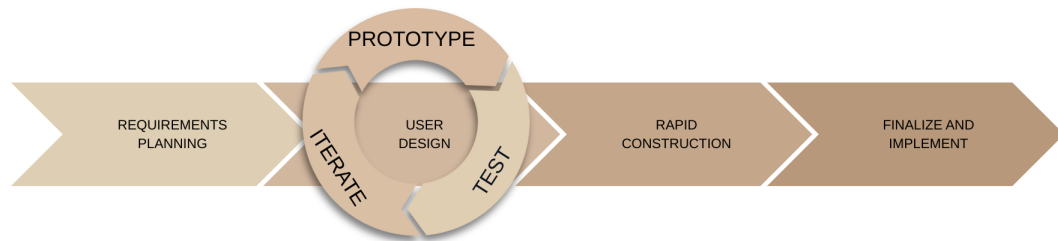


Figure 6: Development model

Requirements Planning: During this phase, one of the researchers visited A.D Soreta Electronics Enterprises in Sorsogon City to meet with the owner and conduct an initial interview on the shop’s current workflows and challenges, After that due to a limited time, a follow-up questionnaire was later given via google forms. After outlining the plans, the said researcher communicated and collaborated with the client regularly to validate the system’s objectives and ensure alignment with their operational needs.

User Design: This stage focused on visualizing the core system components and user interactions. The development team designed diagrams such as Use Case Diagrams, Entity-Relationship Diagrams, and Activity Diagrams to outline the behavior and structure of the system. These tools facilitated better communication between the developers and stakeholders, allowing for shared understanding of how customers would interact with the DIY guide, how appointments would be handled, and how

administrators would manage technician assignments. The user interface design was also prototyped to ensure responsiveness across devices.

Rapid Construction: Once the requirements and visual models were validated, development commenced using the WordPress platform. Plugins were integrated to support appointment booking, user management, and content display. Custom code was applied as needed to tailor functionalities to the specific workflow of A.D. Soreta Electronics Enterprises. The troubleshooting guide was built using structured content to guide customers through common issues, while the booking system was configured to allow customers to set, reschedule, or cancel appointments. Administrator notifications and technician assignment procedures were also implemented during this phase.

Finalize and Implement: After construction, the system underwent thorough testing to ensure functionality, usability, and performance. Testing scenarios were designed to replicate real-world use, from customer self-troubleshooting to successful appointment scheduling and admin technician assignment. Bugs and usability issues were identified and resolved. Once finalized, the system was deployed and made operational, marking the transition from manual to semi-automated service handling at A.D. Soreta Electronics Enterprises. The system's effectiveness will be assessed based on the ISO/IEC 25010 model, focusing on functional suitability, usability, and performance efficiency.

4.4 Development Approach

In developing the system, the developers used the bottom-up development approach, which complemented the Rapid Application Development (RAD) model applied in this project. This approach enabled the team to focus on building the system from its most basic components—such as service request forms, technician assignments, and scheduling—before combining them into the complete Online Service Management System. By breaking the system down into smaller modules, the team was able to thoroughly develop and test each part, addressing issues early in the process and ensuring each feature met the intended requirements. The RAD model allowed for continuous stakeholder feedback, which guided refinements and ensured the system aligned with the operational goals of A.D. Soreta Electronics Enterprises. After individually verifying each module, the developers integrated them into a unified and functional system that supports secure login, technician assignment, appointment scheduling, and notifications—delivered through web and mobile platforms.

4.5 Development Tools

The following development tools and technologies were utilized in creating and deploying the system:

4.5.1 Front End Development Tools

WordPress is an open-source content management system (CMS) designed to create and manage websites through themes, plugins, and an intuitive admin panel [30]. WordPress served as the primary tool for building the system's front-end. Themes were used to structure the layout and design, while plugins provided specific functionalities like appointment scheduling and troubleshooting guides. WordPress allowed the team to easily configure and customize the website without extensive coding knowledge.

Figma is a web-based design and prototyping tool that enables real-time collaboration for designing user interfaces, wireframes, and prototypes [31]. Figma was used to design the system's user interface (UI) and create interactive prototypes, allowing the development team to visualize user flows and layouts before building the site in WordPress. It ensured that the design was both user-friendly and aligned with the system's goals.

4.5.2 Back end Development Tools

XAMPP is a free and open-source cross-platform web server solution that includes Apache, MySQL, PHP, and Perl, facilitating local development environments [32]. XAMPP was employed to simulate a live server environment locally, where the WordPress system could be developed, configured, and tested before deployment. It provided the necessary tools (Apache for server functionality, MySQL for databases, PHP for server-side scripting) for local testing.

MySQL is a relational database management system (RDBMS) used to store, organize, and retrieve structured data in a relational format [33]. MySQL was used to manage the system's database, including customer information, appointment records, and plugin settings. It was accessed via phpMyAdmin within the XAMPP environment to ensure efficient data storage and retrieval.

4.5.3 Documentation Tool

Overleaf (LaTeX) is an online LaTeX editor that facilitates the writing and collaboration of technical documents, with features for precise formatting,

mathematical notation, and references [34]. Overleaf was used for the creation, formatting, and compilation of the capstone project’s technical documentation. This tool provided the necessary features to format the document in IEEE style, include diagrams, tables, and references, and ensure consistency in academic writing.

4.6 Schedule and Timeline

A timeline and schedule are basic yet important aspects of any planning, whether it be a project or an event, as emphasized in project management standards [35]. A schedule maps out the course of action for a particular plan by providing an orderly list of the activities involved in it. As a result, the sequence in which the activities, along with the resources required to finish them, are to be carried out is identified. On the contrary, a timeline is an outline that describes the activities involved and how they relate to one another over a span of time. This timeline fosters more effective teamwork as everyone is on the same wavelength, problem definition, proposed solutions. Equally important is designing a schedule and timeline for a project, computation of time and allowances for risk/events, such as dependencies, and setting of organizational guides of a strategic plan.

Task ID	Task Description	Start Date	End Date	Duration
PLANNING PHASE				
T1	Look for potential client	01/27/25	02/11/25	11d
T2	Contact the client and schedule a meeting	02/09/25	02/14/25	6d
T3	Explain the benefits of the system and understand their business	02/15/25	02/18/25	3d
T4	Observe how their business operates	02/19/25	02/22/25	4d
T5	Think for possible solutions	02/23/25	02/26/25	4d
T6	Define scope and objectives	03/01/25	03/05/25	5d
T7	Think for possible title of the project	03/06/25	03/07/25	2d
T8	Present the title	03/11/25	03/12/25	2d
REQUIREMENTS ANALYSIS PHASE				
T9	Study how the business works in detail	03/13/25	03/21/25	7d
T10	Ask the client what they need from the system	03/22/25	03/26/25	5d
T11	Identify possible challenges and limitations	03/27/25	04/03/25	6d
T12	Confirm and analyze what needs to be included	04/04/25	04/04/25	1d

SYSTEM DESIGN PHASE				
T13	Learn the possible tools and technologies to be used	04/11/25	04/21/25	7d
T14	Design system architecture and database structure	04/22/25	04/30/25	7d
T15	Create wireframes and mock-ups for UI/UX design	04/29/25	05/05/25	7d
T16	Present to the client	05/06/25	05/08/25	3d
T17	Update changes	05/13/25	05/19/25	5d
T18	Design database structure	06/02/25	06/06/25	5d
IMPLEMENTATION PHASE				
T19	Set up tools needed for development	05/31/25	06/08/25	8d
T20	Implement database and back-end functionalities	06/09/25	06/26/25	11d
T21	Develop user interface and front-end components	06/28/25	07/04/25	11d
T22	Integrate system modules and functionalities	07/05/25	07/18/25	10d
T23	Conduct unit testing on individual components	07/18/25	07/28/25	7d
TESTING PHASE				
T24	Check if all features are working properly	07/29/25	08/08/25	9d
T25	Fix any errors or issues found in the system	08/09/25	08/16/25	8d
T26	Let the client test the system and give feedback	08/17/25	08/25/25	9d
T27	Make improvements based on feedback	08/26/25	09/11/25	12d
DEPLOYMENT, MAINTENANCE AND SUPPORT PHASE				
T28	Monitor and update the system as needed	09/14/25	09/25/25	11d
T29	Help client with any system issues	09/27/25	10/06/25	8d
T30	Improve the system based on user experience	10/07/25	10/18/25	10d
T31	Write final report about the project	10/19/25	11/05/25	14d



Figure 7:
Gantt Chart

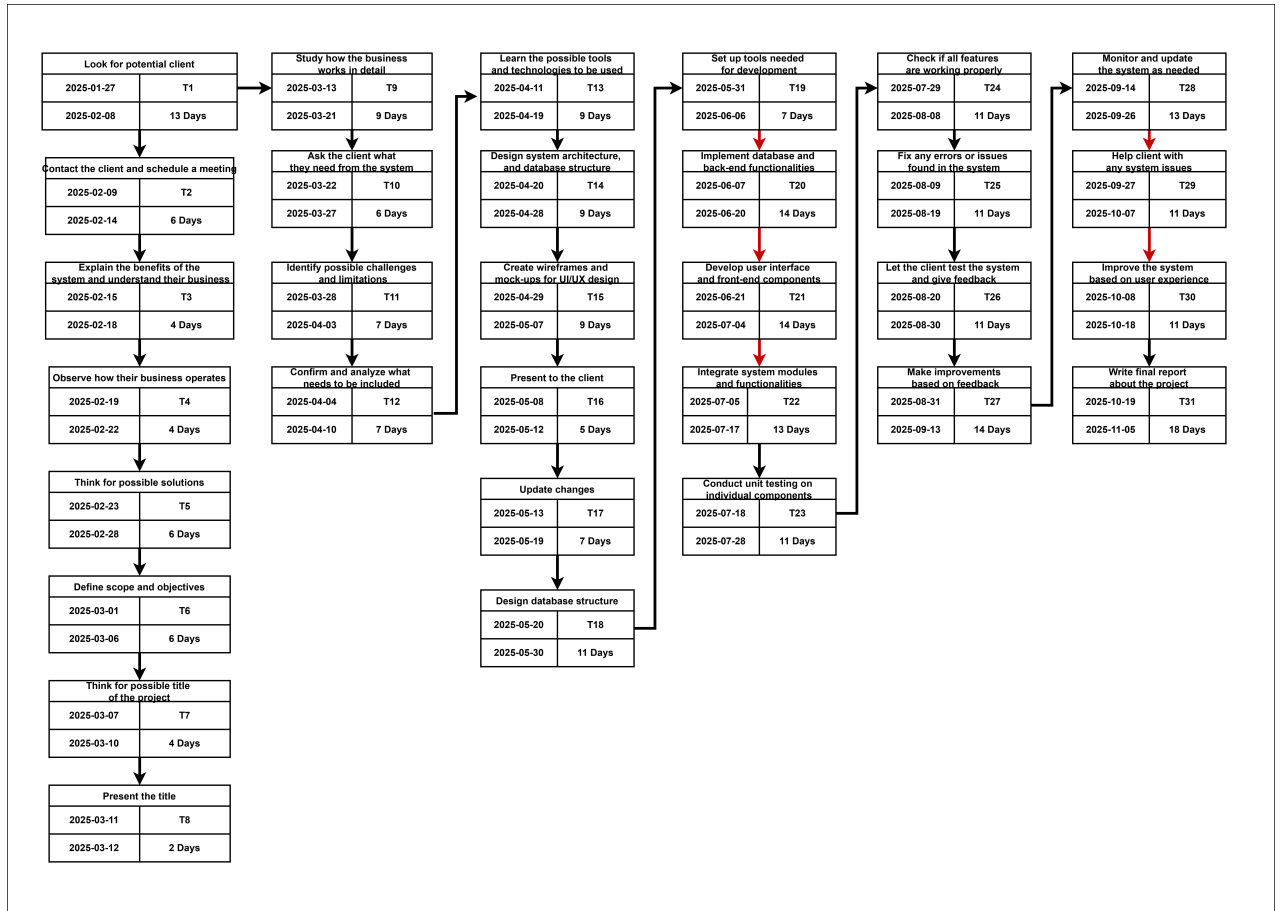


Figure 8: Critical Path Analysis

4.7 Responsibilities

The designated System analysts in our team are Lirah Abaño and Nailia Dipantar . The role of analyst in A.D Soreta is responsible for analyzing the requirements and translating them into technical specifications. Creating diagrams to show users how a product works for the satisfaction of the users.

The designated Technical writer in our team is Zydric Gealone, Technical writers also called technical communicators, prepare instruction manuals, how-to guides, journal articles, requirement information, and other supporting documents to communicate complex and technical information more easily.

The designated Programmer in our team is John Robert Gaufo responsible for writing, testing, debugging, and maintaining the instructions that computers follow to perform tasks. UI/UX designer requirements into computer code, ensuring the software functions correctly and efficiently.

4.8 Budget and Cost Management

This phase contains the detailed examination of the financial aspects associated with the development and operation of the proposed system. This includes the Proponent's budget and cost management in materials and supplies, the services and hosting expenses, and the estimated budget of the proposed system.

4.8.1 Proponent's budget and cost management

The table will outline all the expenses on materials and supplies of the researcher. The total amount of the expenses is Php 740.00.

Table 7: Materials and Supplies

Item Name	Quantity	Price	Amount
Bond paper (long/short)	1 ream	Php 300.00	Php 250.00
Ink (Epson L3210 ink 003)	1 pc	Php 350.00	Php 350.00
White Folder	5 pcs	Php 7.00	Php 140.00
Total			Php 740.00

The Table shown above outlines all the expenses on materials and supplies of the researchers. The total amount of the expenses is PHP 740.00.

4.8.2 Services and Hosting Budget

Table 8: Service Expenses

Service	Monthly Fee	Amount
Internet Connection (BTIT)	Php 1,200.00	Php 12,000.00
WordPress Plugin Ecosystem	Free	Free
Total		Php 12,000.00

In Table 1 and 2, the services hosting budget that the proponents needed are listed. The internet connection, which has a total amount of 1,200 was calculated for a 10 months period of usage. This will be used for gathering information online, collaborating with the development team, and hosting. The researcher will also use the WordPress Plugin Ecosystem as a free and open-source content management (CMS) that lets anyone build and manage websites easily. In summary, the total estimated budget for the Online Service Management System for A.D. Soreta Electronics Enterprises with knowledge-based Technician Assignment project includes Php 740.00 for materials and supplies (Bond paper, ink and folder) and Php Php 12,000.00 for internet connection for 10 months, the overall project expenses is Php 12,678. This Budget ensures the necessary resources for documentation, collaboration, and system deployment through project duration.

4.9 Verification and Validation

The project's development requires validation and verification in order to confirm that the requirements are appropriately met by the system, and to confirm whether the system fulfills the client's expectations and demands. The following topics are covered below:

4.9.1 Verification

The researchers make sure that the design will be put into use in accordance with the guidelines, the system would be carefully planned, designed, system development, examining, and carrying out several types of testing until it is error-free and is prepared for deployment. Additionally, the proponent would visit or stay in touch with the client periodically to ensure that the system's growth meets the needs of the A.D. Soreta electronics.

4.9.2 Validation

The provided table shows a structured approach for validation using the Likert Scale methodology. The Likert Scale is a commonly used tool in surveys and assessments to measure the level of agreement or disagreement with a statement or set of statements [?]. In this case, the Likert Scale is divided into five categories based on the range calculated using the formula (high value - lowest value) / 5, where the high value is 5 and the lowest value is 1.

Mean	Description
4.21 – 5.00	Strongly Agree
3.21 – 4.20	Agree
2.41 – 3.20	Neither Agree nor Disagree
1.81 – 2.40	Disagree
1.00 – 1.80	Strongly Disagree

This table allows us to test and evaluate the usability of the system using the ISO/IEC 25010 series of 2023 software quality standard model. This methodology aids in precisely identifying the amounts of agreement or disagreement, offering a consistent approach to data evaluation and interpretation pertaining to features of software like Functional Suitability, Efficiency, and Usability.

4.10 Testing

System	Online Service Management System for A.D. Soreta Electronics Enterprises with Knowledge-Based Technician Assignment
Access Requirements	URL: Login Credentials (username, password, user type) Required Customer Information: - Full Name - Address - Contact Number - Brand - Product Type - Troubleshoot Details
Goal of the Test	Confirm that the system meets the specified functionality and design requirements outlined by the client through prototype materials such as videos and Figma designs.
Excluded Features (Out of Scope)	Customer Feedback module may not be included in the current predictive analytics testing phase.
Additional Requirements	Implement email notifications for users upon key events (e.g., request updates, technician assignments).
Testing Steps	Login Verification: - Access the provided URL. - Enter login credentials. - Verify successful login and role-based system access.

Test Category	Details
Functionality Testing	- Test each functionality based on client-approved requirements. - Ensure reliable service management functions. - Optimize resource usage and ensure fast response times. - Ensure interface is intuitive and accessible.
Data Input Validation	- Test all data fields for correct input validation. - Reject invalid inputs with appropriate error feedback.
Usability Testing	- Assess UI design for user-friendliness and efficiency. - Verify that users can complete tasks with minimal guidance.
Maintainability Testing	- Check ease of code and module updates. - Ensure updates do not affect existing functionality.
Compatibility Testing	- Test on various browsers (Chrome, Edge). - Verify proper function across desktops, laptops, and smartphones.
Target Device	Desktop computers, laptops, smartphones
Software Required	Compatible web browsers (Chrome, Microsoft Edge)

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